Otopair Enrichment Pipeline

v9 Roadmap | Updated April 4, 2026 (Post-Meeting Revision)

What Changed

The Apr 4 team meeting (Temur, Waleed, Yassin) introduced four major decisions that reshape the enrichment strategy. These add new tasks and re-prioritize the remaining roadmap.

- **30K-mile data ceiling confirmed.** Services repeat on a ~30K-mile cycle (10K/20K/30K). The pipeline only needs to enrich the first 30K miles per model. No new unique intervals after that.
- **23-service default fallback locked.** When pipeline data is missing or incomplete, the app falls back to a pre-built default schedule covering 23 services (~90% of cars).
- **Chassis-based grouping adopted.** Models sharing a chassis/platform reuse the same core maintenance schedule. Only cosmetic parts (headlights, tail lights) differ. Eliminates redundant enrichment.
- **Cross-brand pattern matching.** Yassin proposed identifying shared engine/service patterns across brands (e.g., G80/M3/X3M share an engine). Builds enrichment data off patterns instead of car-by-car research.

Original Tasks	**19**
New Tasks (from meeting)	**+4**
Total Tasks	**23**
Completed (Week 1)	**11 / 23 (48%)**
Blocked	**2 (external deps)**
Remaining	**10 (Weeks 2-3)**

Full Task Status (Updated)

#	Task	Status	Owner	Week
1	NHTSA VIN Decode Integration	**DONE**	Waleed	Wk 1
2	VDB Advanced VIN Decode V2	**DONE**	Waleed	Wk 1
3	Batch API Pipeline (Tier 2)	**DONE**	Waleed	Wk 1
4	Normalized Schema + Write Layer	**DONE**	Waleed	Wk 1
5	Quotable Labor Wiring (SVS Bridge)	**DONE**	Waleed	Wk 1
5A	VDB Repair Estimates API + Labor Mapping	**DONE**	Waleed	Wk 1
8	Drivetrain Consistency (FWD fluid nulling)	**DONE**	Waleed	Wk 1
9	is_staggered Auto-Computation	**DONE**	Waleed	Wk 1
11	Fill Rate Accuracy Fix	**DONE**	Waleed	Wk 1
12	Tier 2 Resilience (3 fixes)	**DONE**	Waleed	Wk 1
13	Source Registry Logging	**DONE**	Waleed	Wk 1
20	**30K-Mile Data Ceiling Implementation**	**NEW**	Waleed	Wk 2
21	**23-Service Default Fallback**	**NEW**	Waleed	Wk 2
22	**Chassis-Based Grouping Lookup**	**NEW**	Waleed	Wk 2
23	**Cross-Brand Pattern Matching Exploration**	**NEW**	Yassin	Wk 2
6	Oto AI Scope Definition	**BLOCKED**	Waleed	Wk 2
14	Mechanic UI Integration	**BLOCKED**	Daniel	Wk 2
7	Evidence + Consensus System	PENDING	Waleed	Wk 2
10	Empirical Learning Expansion	PENDING	Waleed	Wk 2
15	Monthly Price Refresh Cron	PENDING	Waleed	Wk 2-3
16	Admin Dashboard for Pipeline Health	PENDING	Waleed	Wk 3
17	Content Sanitization	PENDING	Waleed	Wk 3
18	Legacy File Cleanup	PENDING	Waleed	Wk 3
19	Handoff Document Update	PENDING	Waleed	Wk 3

Blue-highlighted rows are new tasks added from the April 4 meeting.

New Tasks — Detail Breakdown

Task 20: 30K-Mile Data Ceiling

Services repeat on a ~30K-mile cycle (10K, 20K, 30K). The pipeline should cap enrichment at the first 30K miles per model. No new unique intervals appear beyond that point. One-off items (spark plugs, timing belts at higher mileage) exist but are outside the core recurring schedule.

- Cap service_intervals enrichment at 30K miles
- Remove any interval data beyond 30K from batch prompts
- Add validation: reject intervals > 30K from AI responses

Task 21: 23-Service Default Fallback

When the pipeline returns no data or incomplete data for a vehicle, the app falls back to a pre-built default interval schedule covering 23 services. Covers ~90% of cars. This is the safety net ensuring every user gets a functional maintenance schedule on day one.

- Define the 23-service default schedule (services + intervals + labor hours)
- Build fallback lookup: if vehicle_config has < threshold fill rate, serve defaults
- Merge rule: partial pipeline data + defaults for missing fields (TBD exact logic)

Task 22: Chassis-Based Grouping Lookup

Models sharing a chassis/platform are treated as the same car for maintenance. If the pipeline has already enriched a vehicle with a matching chassis, skip redundant research and reuse that data. Core services (fluids, oil, filters) stay identical. Only cosmetic parts (headlights, tail lights) may differ.

- Add chassis_code field to vehicle_configs (from VIN positions 4-8 or VDB data)
- Before Tier 2 enrichment: check if a matching chassis_code already exists with complete data
- If match found: clone service_intervals, labor_times, drivetrain_config from existing record
- Flag cosmetic-only parts for separate background update

Task 23: Cross-Brand Pattern Matching (Yassin)

Identify shared engine/service patterns across brands to accelerate enrichment at scale. Example: G80, M3, M4, M2, X3M, X4M all share the S58 engine, so their service schedules are functionally identical. Once you identify the pattern, apply it across brands without per-model research.

- Yassin: connect to Cloud MCP and run exploratory queries by Monday
- Map engine families to service schedule groups (e.g., BMW S58, Toyota 2GR)
- Full cross-data pattern review: next Saturday (Apr 11)

Week 1 Completed Work (Recap)

Tasks 1-4: Core Pipeline Build

NHTSA VIN decode, VDB Advanced VIN Decode V2, Tier 2 Batch API pipeline (Batch 1A/1B + Batch 2 gap fill), normalized schema across 9 tables.

Task 5 + 5A: Quotable Labor Wiring

Bridged v3 labor_times with legacy service_vehicle_specs for booking flow. VDB Repair Estimates API at /repair-estimates/{vin}. Priority chain: empirical > VDB > web_search > training_data. Empirical feedback loop in job_actuals.ts.

Task 8: Drivetrain Consistency

FWD vehicles now null out diff_fluid_type and tc_fluid_type. AWD/4WD correctly sets has_transfer_case.

Task 9: is_staggered

Auto-computed from front vs rear tire size comparison. Pure string comparison, no AI needed.

Task 11: Fill Rate Accuracy

Trim fields expanded to 9, FWD no longer inflates fill rate, service denominators use applicableServices.length.

Task 12: Tier 2 Resilience

Concurrent run prevention (4h safety valve), Batch 1+2 submitBatch try-catch, errored/expired batch request detection.

Task 13: Source Registry Logging

Unregistered domains logged with accuracy stats instead of silently skipped.

Bug Fix: VDB Placeholder Engine Codes

Blocklist of 9 placeholder codes (STDEN, STD, etc). Pipeline falls back through NHTSA, Claude AI, web search, synthetic. Tested: 2022 RAV4 Prime resolved to A25A-FXS.

Assignments from Apr 4 Meeting

Who	What	When
Waleed	Pull VINs from CarGurus and populate Convex DB for pattern analysis	This wk + next
Waleed	Implement chassis-based lookup in enrichment pipeline	Ongoing
Yassin	Connect to Cloud MCP, run exploratory queries on enrichment data	By Monday
All	Connect to the Cloud MCP for direct database querying	ASAP
Temur	Distribute meeting notes to the team	Today
All	Full pattern review session across all collected VIN data	Next Saturday

Open Questions

- Chassis-based part differentiation scope for MVP: how are cosmetic parts (headlights, tail lights) managed for models on the same platform?
- Minimum data volume for pattern analysis: after CarGurus pull, is there enough VIN diversity for meaningful cross-brand queries?
- Default fallback + partial pipeline data interaction: if a chassis match exists but enrichment is incomplete, merge partial data with 23-service defaults or pick one?
- Cloud MCP access: setup instructions and permissions not shared in meeting yet.

Updated Week 2-3 Outlook

Week 2 (priority shift): 30K-Mile Ceiling (Task 20), 23-Service Default Fallback (Task 21), Chassis-Based Grouping (Task 22), Cross-Brand Patterns (Task 23 / Yassin), Evidence + Consensus (Task 7), Empirical Learning (Task 10), Price Refresh Cron (Task 15)

Week 3: Admin Dashboard (Task 16), Content Sanitization (Task 17), Legacy Cleanup (Task 18), Handoff Doc (Task 19)

Next checkpoint: Full pattern review session — next Saturday, April 11